



# MICROSOFT POWER PLATFORMS BOOTCAMP

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Microsoft Power Platforms Bootcamp

## Executive Summary

 **Microsoft Power Platform** is a comprehensive low-code ecosystem designed to help organizations rapidly build applications, automate business processes, analyze data, create secure external experiences, and introduce intelligent AI capabilities across their operations.  At the center of the platform is **Microsoft Dataverse**, which provides a secure, structured, and scalable data foundation for storing and managing business information, relationships, security roles, and application logic.  **Power Apps** enables organizations to create responsive Canvas Apps and data-driven Model-Driven Apps that replace spreadsheets, manual processes, and disconnected legacy solutions with modern digital experiences.  **Power Automate** extends these applications through automated, instant, scheduled, approval, integration, and desktop automation workflows, allowing information and tasks to move seamlessly between users, systems, and services.  **Power BI** transforms organizational data into interactive reports, dashboards, KPIs, and actionable insights that support informed decision-making.  **Power Pages** enables organizations to securely expose selected business processes and Dataverse information to external users such as customers, partners, vendors, and citizens through professional websites.  **Microsoft Copilot Studio** adds conversational and AI-driven experiences, allowing organizations to build intelligent agents that can answer questions, perform actions, interact with enterprise data, and participate in automated business processes.  Through hundreds of connectors and integration capabilities, Power Platform can work with Microsoft 365, Dynamics 365, Azure, SharePoint, Teams, SQL databases, APIs, and many third-party services, helping organizations connect previously isolated systems into unified business solutions.  Enterprise capabilities such as role-based security, governance, environments, Data Loss Prevention policies, auditing, monitoring, managed solutions, testing, and Application Lifecycle Management help organizations move from simple prototypes to controlled production deployments.  Rather than viewing each Power Platform product as an isolated tool, organizations gain the greatest value by combining them into an integrated architecture where **Dataverse manages trusted business data, Power Apps provides the user experience, Power Automate orchestrates processes and integrations, Power BI delivers insight, Power Pages extends services externally, and Copilot Studio introduces intelligent interaction and automation.**  Together, these capabilities enable both professional developers and business users to collaborate on solutions that reduce repetitive work, modernize legacy processes, improve data visibility, strengthen operational consistency, and accelerate digital transformation.

## Microsoft Power Platform Bootcamp Roadmap

### 32-Day Hands-On Learning Journey

#### Goal of the Bootcamp

By the end of this bootcamp, students will be able to design, build, automate, secure, analyze, and deploy real business solutions using:

-  Microsoft Dataverse
-  Power Apps
-  Power Automate
-  Power BI
-  Power Pages
-  Copilot Studio
-  Solutions & ALM
-  Governance, Testing & Production Readiness

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#### Month 1 — Foundation to App Building

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##### Week 1 — Power Platform Foundations

*Build the core understanding of the ecosystem*

##### Day 1 — Introduction to Microsoft Power Platform

Learn what Power Platform is, why it exists, and how the products work together.

Deliverable:  Power Platform ecosystem understanding

##### Day 2 — Microsoft Dataverse Fundamentals

Understand tables, columns, rows, relationships, and why Dataverse is the heart of business apps.

Deliverable:  First Dataverse tables

##### Day 3 — Dataverse Data Modeling & Relationships

Go deeper into one-to-many, many-to-many, lookup columns, and data design thinking.

Deliverable: ✅ Relational data model

#### Day 4 — 🔒 Dataverse Security & Business Rules

Learn validation, security roles, record ownership, and basic business rules.

Deliverable: ✅ Secure and structured data foundation

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### 🎨 Week 2 — Power Apps Canvas Apps

*Learn to build interactive apps*

#### Day 5 — 📱 Power Apps Canvas App Fundamentals

Create your first Canvas App and understand screens, controls, properties, and formulas.

Deliverable: ✅ Basic working Canvas App

#### Day 6 — 🔍 Galleries, Forms, Search, Filter & Navigation

Build data-driven app experiences using galleries, forms, search, filter, and navigation.

Deliverable: ✅ Browse-detail-edit style app

#### Day 7 — 🧠 Power Fx Fundamentals & Application Logic

Learn formulas, variables, collections, patching, conditions, and app logic.

Deliverable: ✅ Smarter app behavior

#### Day 8 — 🏗️ Complete Canvas App Architecture & Production Readiness

Understand app structure, state management, validation, responsiveness, and app quality.

Deliverable: ✅ Professional Canvas App structure

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### 🏢 Week 3 — Advanced App Building

*Move from beginner to solution builder*

#### Day 9 — 🧱 Advanced Canvas App Architecture

Learn scalable design patterns, reusable components, and cleaner app structure.

Deliverable:  Better app architecture

Day 10 —  Model-Driven Apps Fundamentals

Understand how Model-Driven Apps work and when to use them.

Deliverable:  First Model-Driven App

Day 11 —  Tables, Views, Forms & App Navigation in Model-Driven Apps

Customize forms, views, navigation, and business experiences.

Deliverable:  Business-ready admin app

Day 12 —  Business Process Logic in Model-Driven Apps

Work with business process flows, rules, and data-driven experiences.

Deliverable:  Structured business process experience

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 Week 4 — Power Automate Fundamentals

*Add automation to your solutions*

Day 13 —  Introduction to Power Automate

Understand triggers, actions, cloud flows, and automation concepts.

Deliverable:  First automated workflow

Day 14 —  Flow Logic, Variables, Conditions & Loops

Learn branching, looping, variables, data operations, and flow structure.

Deliverable:  More intelligent flow logic

Day 15 —  Approval Workflows

Build approval processes using built-in approval actions and status updates.

Deliverable:  Working approval process

Day 16 —  Error Handling, Resilience & Production Flow Design

Learn scopes, retry, Run After, logging, and failure handling.

Deliverable:  Reliable enterprise-style flow

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 Month 2 — Analytics, Portals, AI, Governance & Capstone

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## Week 5 — Power BI Fundamentals

*Turn business data into insights*

### Day 17 — Power BI Fundamentals

Understand reports, dashboards, datasets, and the Power BI workflow.

Deliverable:  First Power BI report

### Day 18 — Power Query, Data Modeling & Star Schema

Learn data cleaning, transformation, facts, dimensions, and model design.

Deliverable:  Clean semantic model

### Day 19 — DAX, Filter Context & Time Intelligence

Build measures, calculations, and time-based analysis using DAX.

Deliverable:  Analytical measures and KPIs

### Day 20 — Power BI Service, Publishing & Governance

Learn publishing, refresh, sharing, workspaces, and report governance.

Deliverable:  Published BI solution

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## Week 6 — Power Pages & AI

*Extend solutions outside the organization and introduce AI*

### Day 21 — Power Pages Fundamentals

Learn how to build external websites connected to Dataverse.

Deliverable:  Starter Power Pages site

### Day 22 — Power Pages Security

Understand contacts, authentication, web roles, page permissions, and table permissions.

Deliverable:  Secure portal architecture

### Day 23 — Copilot Studio Fundamentals

Learn how to build AI-powered copilots and conversational experiences.

Deliverable:  First Copilot/agent

## Day 24 — 🧠 Agent Actions, Orchestration & Lifecycle

Connect copilots to actions, flows, and business data.

Deliverable: ✅ Functional business assistant

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## 🔥 Week 7 — Integration, Governance & ALM

*Make solutions enterprise-ready*

### Day 25 — 🔗 APIs & Custom Connectors

Learn REST APIs, JSON, authentication, and custom connectors.

Deliverable: ✅ Connected external service

### Day 26 — 🛡️ Power Platform Governance

Understand environments, policies, Managed Environments, security, and platform control.

Deliverable: ✅ Governance strategy understanding

### Day 27 — 📦 Solutions & Application Lifecycle Management (ALM)

Learn solution packaging, connection references, environment variables, and deployment.

Deliverable: ✅ Solution-aware deployment design

### Day 28 — 🧪 Testing, Troubleshooting & Production Readiness

Learn functional testing, UAT, troubleshooting, monitoring, and release quality checks.

Deliverable: ✅ Production-readiness checklist

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## 🏆 Week 8 — Final Capstone Project

*Bring everything together into a real-world solution*

### Day 29 — 🏗️ Capstone Architecture & Dataverse Design

Design the business problem, solution architecture, tables, relationships, and security model.

Deliverable: ✅ Final ERD + architecture diagram

### Day 30 — 🧩 Enterprise App Build

Build the main application experience using Canvas App and/or Model-Driven App.

Deliverable:  Working business application

Day 31 —  Automation, Admin Experience & Analytics

Add Power Automate flows, admin tools, and Power BI reporting.

Deliverable:  End-to-end working solution

Day 32 —  Final Testing, Deployment & Solution Presentation

Test, document, package, and present the final project.

Deliverable:  Final capstone demo + deployment-ready solution

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## Learning Journey Summary

### Phase 1 — Foundations

You start by understanding:

-  The Power Platform ecosystem
-  Dataverse
-  Security
-  Data modeling

### Phase 2 — App Development

Then you learn:

-  Canvas Apps
-  Model-Driven Apps
-  Power Fx
-  User experience and architecture

### Phase 3 — Automation

Then you add:

-  Flows

-  Approvals
-  Error handling
-  Resilience

#### Phase 4 — Analytics & Extensions

Then you expand into:

-  Power BI
-  Power Pages
-  Copilot Studio
-  APIs and integration

#### Phase 5 — Enterprise Readiness

Finally, you learn:

-  Governance
-  ALM
-  Testing
-  Deployment
-  Capstone delivery

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#### Final Outcome of the Bootcamp

By the end of the 32 days, students should be able to:

Understand business requirements



Design a data model



Build apps



**Automate business processes**



**Secure the solution**



**Create analytics**



**Package and deploy**



**Present a complete business solution**

**They should be able to think not only like:**

-  App makers
-  Flow builders
-  Report creators

**but also like:**

 **Power Platform Solution Architects**

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## **Final Bootcamp Vision**

**This bootcamp takes students from:**

**Beginner**



**Builder**



**Solution Developer**



**Professional Power Platform Practitioner**

## Microsoft Power Platform Bootcamp — 32-Day Lab Roadmap

**Approach:** Hands-on Labs · Progressive Projects · Enterprise Capstone

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### WEEK 1 — Dataverse Foundations

#### Day 1 — Introduction to Power Platform

**Deliverable:** Employee Leave Management solution architecture

#### Day 2 — Dataverse Fundamentals

**Deliverable:** Department, Position & Employee tables

#### Day 3 — Relationships & Business Logic

**Deliverable:** Complete Leave Management data model

#### Day 4 — Security, Environments & Solutions

**Deliverable:** Security roles + DEV/TEST/PROD solution structure

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### WEEK 2 — Canvas Apps

#### Day 5 — Canvas App Fundamentals

**Deliverable:** Employee Leave App v0.1

#### Day 6 — Galleries, Search, Filters & Forms

**Deliverable:** Employee Leave App v0.2

#### Day 7 — Power Fx Fundamentals

**Deliverable:** Employee Leave App v0.3 with business logic

#### Day 8 — Complete Canvas App

**Deliverable:** Employee Leave Management App v1.0

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### WEEK 3 — Advanced Power Apps

#### Day 9 — Advanced Canvas Architecture

**Deliverable:** IT Help Desk App v0.1

#### Day 10 — Advanced Power Fx & Data Manipulation

**Deliverable:** IT Help Desk App v0.2

#### Day 11 — Data Integration & Connectors

**Deliverable:** Microsoft 365, SharePoint & API integrations

## **Day 12 — Model-Driven Apps**

**Deliverable:** IT Help Desk Administration App v1.0

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## **WEEK 4 — Power Automate**

### **Day 13 — Power Automate Fundamentals**

**Deliverable:** Ticket notification & confirmation flows

### **Day 14 — Approval Workflows**

**Deliverable:** Employee Leave Manager Approval Flow v1.0

### **Day 15 — Advanced Flow Logic & Error Handling**

**Deliverable:** Resilient Leave Approval Flow v1.1

### **Day 16 — End-to-End Automation & Monitoring**

**Deliverable:** Production-ready automation package

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## **WEEK 5 — Power BI**

### **Day 17 — Power BI Fundamentals**

**Deliverable:** HR Leave Analytics v0.1

### **Day 18 — Power Query & Star Schema Modeling**

**Deliverable:** HR Analytics semantic model v0.2

### **Day 19 — DAX & Business Metrics**

**Deliverable:** HR Leave Analytics v0.3 with KPIs

### **Day 20 — Complete Power BI Reporting**

**Deliverable:** HR Leave Analytics Report v1.0

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## **WEEK 6 — Power Pages & Copilot Studio**

### **Day 21 — Power Pages Fundamentals**

**Deliverable:** Vendor Registration Portal v0.1

### **Day 22 — Power Pages Security & Authorization**

**Deliverable:** Secure Vendor Portal v1.0 baseline

### **Day 23 — Copilot Studio Fundamentals**

**Deliverable:** HR Support Agent v0.1

## **Day 24 — Agent Actions & Integrations**

**Deliverable:** HR Agent with authenticated status/actions

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## **WEEK 7 — Enterprise Integration, Governance & ALM**

### **Day 25 — APIs & Custom Connectors**

**Deliverable:** Enterprise Custom Connector integration

### **Day 26 — Power Platform Governance**

**Deliverable:** Governance, environment & Data Policy framework

### **Day 27 — Solutions & Application Lifecycle Management**

**Deliverable:** Managed DEV → TEST deployment process

### **Day 28 — Enterprise Testing & Troubleshooting**

**Deliverable:** Production Readiness Assessment v1.0

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## **WEEK 8 — Final Enterprise Capstone**

### **Procurement & Purchase Request Management System**

#### **Day 29 — Capstone Architecture & Dataverse Design**

**Deliverable:** Procurement Solution Architecture v0.1

#### **Day 30 — Canvas App & Shopping Cart**

**Deliverable:** Employee Purchase Request App v0.1

#### **Day 31 — Approvals, Administration & Analytics**

**Deliverable:** Procurement Management Release Candidate

#### **Day 32 — Testing, Deployment & Presentation**

**Deliverable:** Procurement & Purchase Request Management System v1.0

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## **Learning Journey**

DATaverse



CANVAS APPS



ADVANCED POWER APPS

↓  
POWER AUTOMATE  
↓  
POWER BI  
↓  
POWER PAGES + COPILOT  
↓  
APIs + GOVERNANCE + ALM  
↓  
ENTERPRISE CAPSTONE

### Final Outcome

By the end of the 32-day bootcamp, students will be able to:

**Design → Build → Automate → Analyze → Secure → Govern → Test → Deploy**

complete Microsoft Power Platform business solutions.

| Day Main Topic |                                                | Key Deliverable                                   |
|----------------|------------------------------------------------|---------------------------------------------------|
| 1              | Introduction to Microsoft Power Platform       | Employee Leave Management solution architecture   |
| 2              | Dataverse Fundamentals                         | Department, Position, and Employee tables         |
| 3              | Dataverse Relationships & Business Logic       | Complete Leave Management data model              |
| 4              | Dataverse Security, Environments & Solutions   | Security roles + DEV/TEST/PROD solution structure |
| 5              | Canvas App Fundamentals                        | Employee Leave App v0.1                           |
| 6              | Galleries, Search, Filters, Forms & Navigation | Employee Leave App v0.2                           |

| <b>Day</b> | <b>Main Topic</b>                     | <b>Key Deliverable</b>                          |
|------------|---------------------------------------|-------------------------------------------------|
| <b>7</b>   | Power Fx Fundamentals                 | Employee Leave App v0.3 with business logic     |
| <b>8</b>   | Complete Canvas App Design            | Employee Leave Management App v1.0              |
| <b>9</b>   | Advanced Canvas App Architecture      | IT Help Desk App v0.1                           |
| <b>10</b>  | Advanced Power Fx & Data Manipulation | IT Help Desk App v0.2                           |
| <b>11</b>  | Data Integration & Connectors         | Microsoft 365, SharePoint, and API integrations |
| <b>12</b>  | Model-Driven Apps                     | IT Help Desk Administration App v1.0            |
| <b>13</b>  | Power Automate Fundamentals           | Ticket notification and confirmation flows      |
| <b>14</b>  | Power Automate Approvals              | Employee Leave Manager Approval Flow v1.0       |
| <b>15</b>  | Advanced Flow Logic & Error Handling  | Resilient Leave Approval Flow v1.1              |
| <b>16</b>  | End-to-End Automation & Monitoring    | Production-ready automation package             |
| <b>17</b>  | Power BI Fundamentals                 | HR Leave Analytics v0.1                         |
| <b>18</b>  | Power Query & Star Schema Modeling    | HR Leave Analytics semantic model v0.2          |
| <b>19</b>  | DAX Measures & Filter Context         | HR Leave Analytics v0.3 with business KPIs      |
| <b>20</b>  | Complete Power BI Reporting           | HR Leave Analytics Report v1.0                  |
| <b>21</b>  | Power Pages Fundamentals              | Vendor Registration Portal v0.1                 |
| <b>22</b>  | Power Pages Security & Authorization  | Secure Vendor Portal v1.0 baseline              |
| <b>23</b>  | Copilot Studio Fundamentals           | HR Support Agent v0.1                           |

| <b>Day</b> | <b>Main Topic</b>                               | <b>Key Deliverable</b>                                   |
|------------|-------------------------------------------------|----------------------------------------------------------|
| <b>24</b>  | Copilot Studio Actions & Integrations           | HR Agent with authenticated actions/status retrieval     |
| <b>25</b>  | APIs & Custom Connectors                        | Working enterprise Custom Connector integration          |
| <b>26</b>  | Power Platform Governance                       | Governance, environment, and Data Policy framework       |
| <b>27</b>  | Solutions & Application Lifecycle Management    | Managed ALM pipeline from DEV to downstream environments |
| <b>28</b>  | Enterprise Testing & Troubleshooting            | Production Readiness Assessment v1.0                     |
| <b>29</b>  | Capstone Architecture & Dataverse Design        | Procurement Solution Architecture v0.1                   |
| <b>30</b>  | Capstone Canvas App & Shopping Cart             | Employee Purchase Request App v0.1                       |
| <b>31</b>  | Capstone Approvals, Model-Driven App & Power BI | Procurement Management Release Candidate                 |
| <b>32</b>  | Final Testing, Deployment & Presentation        | Procurement & Purchase Request Management System v1.0    |